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MILITARY SPECIFICATION

ASPIRIN, ALUMINUM HYDROXIDE GEL, DRIED, AND MAGNESIUM HYDROXIDE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Aspirin, Aluminum Hydroxide Gel, Dried, and Magnesium Hydroxide Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Aspirin, Aluminum Hydroxide Gel, Dried, and Magnesium Hydroxide Tablets and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type I.

S5.3 The following is added to this paragraph:

Shall be tablets containing 600 mg of Aspirin, 150 mg of Aluminum Hydroxide Gel, Dried, and 150 mg of Magnesium Hydroxide per tablet, within the designated assay limits for the tablet.

S6.4.9.1 Tablets shall disintegrate in not more than 5 minutes when tested according to the USP Disintegration Test for uncoated tablets.

3.2 Assay of the finished tablets.

3.2.1 Aspirin. Shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of aspirin when determined as specified in the USP for Aspirin Tablets.

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3.2.2 Aluminum hydroxide gel, dried and magnesium hydroxide. Shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of aluminum hydroxide gel, dried and magnesium hydroxide when determined as specified in 4.5.1 and 4.5.2.

3.2.3 Free salicylic acid (non-aspirin salicylates) Shall assay to contain not more than 3.0 percent free salicylic acid based on the labeled aspirin content when determined as specified in the USP for Aspirin Tablets (buffered).

3.3 Identification of active ingredient on the finished tablet.

3.3.1 Aspirin. Shall comply with Identification Test A and B in the USP monograph for Aspirin Tablets.

3.3.2 Aluminum hydroxide gel, dried and magnesium hydroxide. The identification of aluminum hydroxide gel, dried and magnesium hydroxide shall be determined as specified in 4.5.3.1 and 4.5.3.2.

3.4 Ingredients.

3.4.1 Aspirin and aluminum hydroxide gel, dried. The aspirin and aluminum hydroxide gel, dried, used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.4.2 Magnesium hydroxide. The magnesium hydroxide used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

3.5 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.6 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.5 Tests - assays. Treat the tablets according to the following procedure prior to assaying for aluminum hydroxide gel, dried and magnesium hydroxide.

> Transfer 20 tablets to a 500 ml erlenmeyer flask, add about 150 ml of diluted hydrochloric acid USP, heat on a steam bath for 15 to 20 minutes and allow to cool. Add about 150 ml of purified water and swirl to resuspend the precipitate. Filter the suspension through Whatman No. 41 filter paper or equivalent, with the aid of a buchner funnel or equivalent and light suction. Rinse the erlenmeyer flask with four 25 ml portions of diluted hydrochloric acid USP and use the rinses to wash the precipitate. Quantitatively transfer the combined filtrates into a 1000 ml volumetric flask using purified water to facilitate the transfer, dilute to volume with purified water and mix. Filter the solution through Whatman No. 41 filter paper or equivalent to remove residual particulate matter and collect about 200 ml of filtrate. Use this clear filtrate in the assay for aluminum hydroxide gel, dried, and magnesium hydroxide.

4.5.1 Assay for aluminum hydroxide gel, dried. Pipet 20 ml of the clear filtrate and follow the procedure as directed in the assay for aluminum oxide in the USP monograph for Alumina and Magnesia Oral Suspension, beginning with, "... into a 250 ml beaker, ...". Calculate the quantity of aluminum oxide per tablet.

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4.5.2 Assay for magnesium hydroxide. Pipet 20 ml of the clear filtrate and follow the procedure as directed in the assay for magnesium hydroxide in the USP monograph for Alumina and Magnesia Oral Suspension, beginning with "... into a 400 ml beaker ...". Calculate the quantity of magnesium hydroxide per tablet.

4.5.3 Identification of aluminum hydroxide gel, dried and magnesium hydroxide. Treat the tablets according to the following procedure prior to the identification of the aluminum and magnesium.

Transfer 20 tablets to a 250 ml beaker, add about 75 ml of diluted hydrochloric acid, USP, heat on a steam bath for 15 to 20 minutes and allow to cool. Add about 50 ml of purified water and swirl to resuspend the precipitate. Filter the suspension through Whatman No. 41 filter paper or equivalent. To 20 ml of the filtrate add ammonia TS dropwise until the solution is just alkaline to litmus paper. Filter the precipitate through Whatman No. 41 filter paper or equivalent; this precipitate is used for identification of aluminum. Save the filtrate which is used for identification of magnesium.

4.5.3.1 Identification for aluminum. Wash the precipitate with about 50 ml of hot ammonium chloride solution (1 in 50). Dissolve the precipitate in 5 ml of diluted hydrochloric acid USP, add 10 ml of purified water, mix and divide the solution into three 5 ml portions. To the first portion add slowly ammonia TS: a precipitate is formed which is not soluble when excess ammonia TS is added. To the second portion add slowly sodium hydroxide TS: a precipitate is formed which dissolves when excess sodium hydroxide TS is added. To the third portion add slowly sodium sulfide TS: a precipitate is formed which dissolves when excess sodium sulfide TS is added.

4.5.3.2 Identification for magnesium. To the filtrate previously saved, add 5 ml of ammonium carbonate TS and mix: no precipitate should form. Add 3 ml of sodium phosphate TS and mix: a white precipitate is formed. Add 10 ml of ammonia TS and mix: the precipitate should remain undissolved.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 100 tablets, as specified, constitutes one unit of issue.

5.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle with a child-proof closure considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Thirty-six (36) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Thirty-six (36) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

* 5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

* 5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

* NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

* 5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

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5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-039-5846	ASPIRIN, ALUMINUM HYDROXIDE GEL, DRIED, AND MAGNESIUM HYDROXIDE TABLETS, 100s

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activity:
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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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